# MUHAMMAD UMER KHAN

**AI Engineer | Agentic AI & RAG | Voice AI & Workflow Automation**

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## SUMMARY

AI Engineer, 2+ years building production LLM systems: agentic RAG assistants, multi-agent backends, voice and WhatsApp agents, CRM automation. Delivered 12+ projects end to end. The work spans fine-tuning to evaluation-gated CI deploys, including white-label SaaS live in production, and every system is red-teamed against OWASP LLM Top 10 with findings published.

## TECHNICAL SKILLS

**Agentic AI:** AI Agents, Multi-Agent Systems, Prompt Engineering, Tool Calling, LangGraph, LangChain, CrewAI
**Fine-Tuning & Evaluation:** LoRA/PEFT, Quantization, LLM-as-Judge, Faithfulness/Relevancy Metrics, DeepEval
**RAG & Vector Search:** Hybrid Search (BM25 + dense), Rank Fusion, Cross-Encoder Reranking, Qdrant, FAISS
**LLM Safety:** Guardrails, Prompt-Injection Defense, Red-Teaming, Threat Modeling, OWASP LLM Top 10, MITRE ATLAS
**MLOps & Deployment:** FastAPI, REST APIs, Docker, GitHub Actions CI/CD, Distributed Tracing, MLflow, LangSmith
**ML & Data Science:** Python, PyTorch, TensorFlow, Scikit-Learn, NLP, Computer Vision, Pandas, NumPy
**Voice & Automation:** Inbound/Outbound Voice Agents, Speech-to-Text, WhatsApp Automation, Vapi, Retell AI, n8n, Make
**Databases & Cloud:** PostgreSQL, MongoDB, SQL, Supabase, AWS, Azure, Git, React, Next.js, Expo (React Native)
**Languages:** English (Professional), Urdu (Native)

## EXPERIENCE

**AI Engineer — Revera Innovations** *Karachi (Hybrid) | Feb 2025 – Present*

- Deployed two vision products to production: a virtual try-on system, and an autonomous skin-analysis agent that detects skin concerns in a user's photo and reasons over live inventory to recommend products.
- Built inbound and outbound AI voice agents on Vapi and Retell AI. One is a dental-clinic receptionist: it books open slots through Cal.com, creates the lead and deal in HubSpot, and alerts the team.
- Automated CRM operations with n8n and Make across HubSpot, GoHighLevel, Airtable and Jotform.
- Main developer since June 2026 of a white-label society-management SaaS live in production (React, Expo, Supabase): sub-admin roles enforced by row-level security, a database-enforced gate blocklist and USB scanner input.

**AI Developer — Bright Solutions** *Karachi (Hybrid) | Dec 2024 – Feb 2025*

- Launched AI voice agents and automated lead-generation systems, now part of daily production operations.

**AI Engineer Intern — SaylaniTech** *Karachi (On-site) | Jun 2024 – Dec 2024*

- Developed LLM and NLP features and shipped them into applications behind production-grade APIs.

**Data Science Intern — Digital Empowerment Network** *Remote | May 2024 – Jun 2024*

- Built predictive models on social-impact datasets, plus the data-cleaning pipelines and reports behind them, in Python.

## PROJECTS

**WhatsApp Hotel-Lead Bot — built at Saylani Tech** *Case Study | 2026*

- Live in production for a travel agency. GPT-5.4 nano proposes a typed lead patch through structured outputs; Python validates and merges it, routes the confirmed lead to the team and relays their reply. 233 tests.

**AskUmer — Hybrid-RAG Conversational Assistant** *GitHub | 2026*

- A LangGraph ReAct agent over hybrid retrieval: Qdrant dense and BM25 sparse search, rank fusion and FlashRank reranking across 898 indexed chunks. An in-memory knowledge graph handles multi-hop queries.
- Hardened with NeMo Guardrails input gating and Portkey failover (gpt-oss-120b → gpt-oss-20b, retries, caching). Deploys are gated by a DeepEval LLM-as-judge CI suite at ≥ 0.7 faithfulness and answer relevancy.

**DineMate — Agentic AI Food-Ordering System** *Live Demo · Security Report | 2025*

- Built a four-agent LangGraph backend (ordering, order management, kitchen, admin) with voice ordering via Whisper ASR and TTS. Role-gated dashboards track revenue, peak hours and top items.
- Red-teamed it across 8 attack categories mapped to OWASP LLM Top 10 and MITRE ATLAS. A critical price-manipulation flaw was fixed with server-side recomputation, and a Llama Prompt Guard 2 classifier now screens input.

**SupportGenie — Dual Fine-Tuned Banking Support Chatbot** *GitHub | 2025*

- LoRA fine-tuned Mistral-7B (17,000 instruction pairs, r=32) and LLaMA-3-8B (1,764 pairs) in 4-bit quantization on one free-tier T4. Final loss: 1.78 / 1.49. FAISS FAQ retrieval answers first, falling through to the models.

## EDUCATION

**B.S. in Computer Science — Sindh Madressatul Islam University** *Karachi | 2023 – 2027 (Expected)*

## CERTIFICATIONS

- DeepLearning.AI — Deep Learning + NLP (2025) · IBM — ML Professional Certificate (2024) · SMIT — AI & Data Science (2025)